Some Common Use Cases of QPUs

- ***Medical and pharmaceutical.*** By reducing the reliance on trial-and-error and increasing efficiency, the target identification, drug design, and toxicity testing might all be far faster, more targeted, and more precise thanks to quantum processors.
- ***Automotive.*** Quantum computing has applications for the automobile sector in research and development, product design, supply-chain management, production, mobility, and traffic control.
- ***Artificial intelligence.*** With regard to applications in artificial intelligence, quantum computers' capacity to analyse enormous data sets, simulate intricate models, and swiftly resolve optimisation issues has attracted interest.
- ***Logistics.*** Optimisation comes easily to quantum computers. Theoretically, a quantum computer might solve a challenging optimisation issue in a matter of minutes that would take a supercomputer years to complete. Hence, it can help tackle difficult logistics challenges.
- ***National security.*** Global governments, including the government of India, are making significant investments in quantum computing research projects, mainly to boost national security.
- ***Defense.*** Quantum computers could be used for a variety of defense-related tasks, including cracking spy codes, simulating battles, and creating better materials for military equipment.

A Customised QPU for Your Application

QuantWare, a Netherlands based company, has declared that it can now provide researchers and other companies in the field with a custom 25-qubit QPU. This new QPU can be delivered, according to QuantWare, in just 30 days! Buyers can select the components they want for the new QPU from a library, and they can also decide how the qubits are connected based on their individual requirements.

applications of quantum computing, and whether you have the funds to facilitate them.

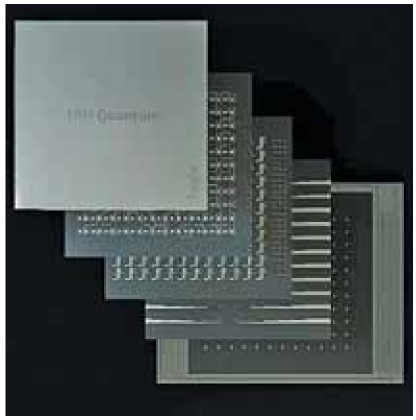
As seen previously in this article, every quantum computer is built using a somewhat distinct hardware design and a multitude of qubit kinds. Each can also have a different architecture. Because of this diversity, certain systems will be more effective at resolving various types of issues than others. Therefore, you must be prepared to consider a variety of factors while selecting a QPU for any application. Here are some pointers that will make that decision easier.

Identifying and analysing the problem. First, you need to identify the problem in your company or organisation that has the potential to be solved by quantum computing. The ideal strategy is to first identify the parameters that most likely indicate quantum opportunities. These parameters include complexity, problem size, potential for more precise outcomes, and potential for quicker turnaround times for results.

You should then analyse the problem in terms of its processing and mathematical computing requirements, then relate those requirements to the capabilities of a variety of QPUs. Refer the box item above to have a look at some common-use cases of QPUs.

Testing the QPU. Before investing, you need to test the system to make sure it meets your requirements and benchmarks. The typical method would be to perform a number of tests for a few qubit operations and compare the findings with those from a conventional (classical) computer model imitating the quantum tech.

Be aware that the output of quantum computers differs from that of classical systems. A quantum computer, for instance, produces a variety of outcomes for optimisation problems that all satisfy the optimisation requirements. Classical systems only produce the best outcome.


The Eagle Chip (Credit: IBM Quantum)

Testing the software. To operate a quantum computer, new code and algorithms must be written. In order to test them, you can either test a single algorithm on a quantum computer or use ready-to-run software.

Important performance metrics of a QPU

In their paper, High-Performance Computing with Quantum Processing Units, researchers Keith A. Britt and Travis S. Humble identify various performance metrics of QPUs, which capture the interplay between system architecture and the quantum parallelism that underlies computational performance. Here, FTQEC stands for fault-tolerant quantum error correction, an error correction technique commonly used in QPUs.

"The comparison of these metrics with each other offers a quantitative means of assessing the value of QPU-enabled systems, but only when they can be related to existing system metrics, e.g., FLOPS, etc," wrote the authors.

India and the quantum race

With strategic partnerships, budding startups, early investments and adoption, India has joined the quantum race and aims to be the quantum hub of the future. "Indian philosophy and scriptures have been talking about